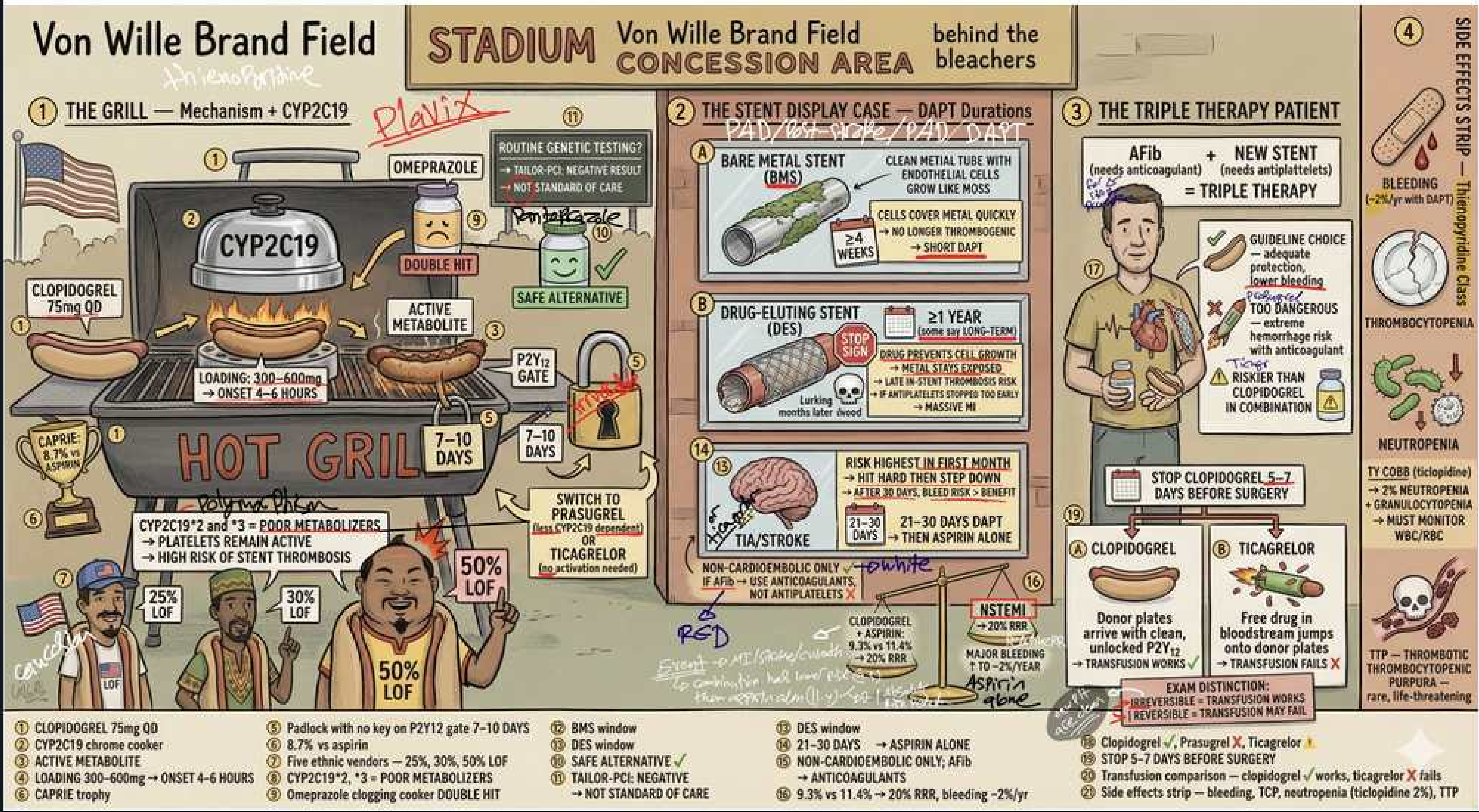
Antiplatelets — Clopidogrel (Plavix, CYP2C19 Grill)



1. CYP2C19 hot grill — prodrug activation

WHAT YOU SEE
A flaming grill labeled CYP2C19 cooking clopidogrel into an 'active metabolite' on a HOT GRILL.

WHAT IT ENCODES
Clopidogrel (Plavix) is an inactive PRODRUG that must be activated by hepatic CYP2C19 to its active thiol metabolite, which then irreversibly blocks P2Y12. Activation is the rate-limiting, variable step.

BOARD TRAP
Because activation depends on CYP2C19, clopidogrel has variable response. Prasugrel and ticagrelor are far less CYP2C19-dependent — the discriminator when a patient fails clopidogrel.

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2. Clopidogrel 75 mg + P2Y12 blockade

WHAT YOU SEE
Clopidogrel 75mg OD label feeding the grill; P2Y12 gate padlocked.

WHAT IT ENCODES
The active metabolite IRREVERSIBLY binds the P2Y12 ADP receptor, preventing ADP-driven amplification and GPIIb/IIIa activation for the platelet lifespan. Maintenance 75 mg daily; loading 300-600 mg.

BOARD TRAP
Clopidogrel is irreversible (like prasugrel) — held 5-7 days pre-op. Ticagrelor, although reversible, is still held ~5 days. Don't assume reversible = no hold.

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3. Loading 300-600 mg, onset 4-6 h

WHAT YOU SEE
Annotation 'LOADING: 300-600mg' and 'ONSET 4-6 HOURS'.

WHAT IT ENCODES
Loading dose 300-600 mg gives a faster onset (still ~2-6 h); maintenance 75 mg daily. The 600 mg load achieves quicker, stronger inhibition than 300 mg for PCI.

BOARD TRAP
Onset is slow (hours) versus prasugrel/ticagrelor (~30 min). For urgent potent inhibition, clopidogrel's delayed onset is a limitation.

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4. Omeprazole — CYP2C19 inhibitor (double hit)

WHAT YOU SEE
Omeprazole bottle with red 'DOUBLE HIT'; esomeprazole crossed out; pantoprazole marked SAFE ALTERNATIVE.

WHAT IT ENCODES
Omeprazole/esomeprazole inhibit CYP2C19 and can blunt clopidogrel activation, theoretically reducing antiplatelet effect. Pantoprazole is the preferred PPI because it inhibits CYP2C19 much less.

BOARD TRAP
If a clopidogrel patient needs a PPI, choose PANTOPRAZOLE — omeprazole/esomeprazole are the CYP2C19-inhibiting 'double hit'. This PPI interaction is a classic board point.

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5. CYP2C19 poor metabolizers

WHAT YOU SEE
Slumped 'CYP2C19 *2 and *3 = POOR METABOLIZERS -> platelets remain active -> high risk of stent thrombosis'; 25%/30%/50% LOF figures.

WHAT IT ENCODES
Loss-of-function alleles (CYP2C19*2, *3) reduce active-metabolite formation, leaving platelets active and raising stent thrombosis/MACE risk. ~25-50% loss of function depending on genotype.

BOARD TRAP
For LOF poor metabolizers, switch to PRASUGREL or ticagrelor (not CYP2C19-dependent) rather than just escalating clopidogrel dose. Routine genotyping is not universally standard.

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6. Routine genetic testing? Not standard

WHAT YOU SEE
Banner 'ROUTINE GENETIC TESTING? -> TAILOR-RX: negative result -> not standard of care'.

WHAT IT ENCODES
Routine pharmacogenomic CYP2C19 testing before clopidogrel is NOT standard of care; selective testing may guide therapy in high-risk PCI but isn't mandated.

BOARD TRAP
Don't answer 'genotype every patient before clopidogrel' — it is not standard of care. Reserve testing for selected high-risk cases.

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7. Bare-metal stent (BMS) — DAPT >=1 month

WHAT YOU SEE
Stent display case A: BARE METAL STENT, 'clean metal, thin with endothelial cells grow over fast, >=4 weeks'.

WHAT IT ENCODES
BMS endothelializes quickly, so the minimum DAPT (aspirin + P2Y12) duration is about 1 month (>=4 weeks). Shorter DAPT is acceptable if bleeding risk is high.

BOARD TRAP
BMS needs SHORTER DAPT (~1 month) than DES. Confusing BMS (4 weeks) with DES (>=6-12 months) DAPT duration is a frequent stent trap.

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8. Drug-eluting stent (DES) — DAPT >=6-12 months

WHAT YOU SEE
Stent case B: DRUG-ELUTING STENT, 'drug prevents cell growth -> delayed endothelialization -> >=1 year'.

WHAT IT ENCODES
DES release antiproliferative drug that delays endothelialization, so they need longer DAPT (typically 6-12 months) to prevent late stent thrombosis. Risk is highest in the first month.

BOARD TRAP
Stopping DAPT early after DES is the dangerous trap — late stent thrombosis. Highest risk is the first month; 'hit hard, then step down'.

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9. Stent thrombosis vs TIA/stroke timing

WHAT YOU SEE

Brain/clot icon: 'risk highest in first month — hit hard then step down; TIA/stroke 21-30 days -> aspirin alone; 21-30 days part -> then aspirin alone'.

WHAT IT ENCODES

Stent thrombosis risk peaks early then declines; after the mandated DAPT window many patients de-escalate to aspirin monotherapy. Timing drives the antiplatelet de-escalation decision.

BOARD TRAP

Premature discontinuation causes catastrophic stent thrombosis; the discriminator is matching DAPT duration to stent type and bleeding risk, not stopping at a fixed arbitrary date.

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10. Triple therapy — AFib + new stent

WHAT YOU SEE

Triple Therapy Patient: AFib (needs anticoagulant) + new stent (needs antiplatelets) = triple therapy; 'too dangerous, shorten, lower bleeding'.

WHAT IT ENCODES

A patient with AFib (needs an oral anticoagulant) who also gets a stent (needs DAPT) may transiently need TRIPLE therapy (OAC + aspirin + P2Y12), but it markedly raises bleeding — minimize duration.

BOARD TRAP

Triple therapy is high-bleed: guideline trend is to drop aspirin early and use OAC + clopidogrel (preferred P2Y12 here, not prasugrel/ticagrelor) to limit bleeding. Prefer clopidogrel in combination.

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11. Stop clopidogrel 5-7 days before surgery

WHAT YOU SEE

'STOP CLOPIDOGREL 5-7 DAYS BEFORE SURGERY'; donor-platelet panel; transfusion fails note.

WHAT IT ENCODES

Because inhibition is irreversible for the platelet lifespan, clopidogrel is held ~5-7 days before elective surgery to allow fresh, uninhibited platelets to regenerate.

BOARD TRAP

Platelet transfusion may help reverse clopidogrel/prasugrel (irreversible) but NOT ticagrelor — circulating ticagrelor inhibits the transfused platelets too. Don't transfuse to reverse ticagrelor.

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12. Clopidogrel vs Ticagrelor (free vs locked)

WHAT YOU SEE

Side-by-side: CLOPIDOGREL (donor platelets arrive clean, unlocked P2Y12) vs TICAGRELOR (free drug in bloodstream jumps onto donor plates).

WHAT IT ENCODES

Clopidogrel/prasugrel covalently and irreversibly block P2Y12 (prodrugs); ticagrelor binds reversibly and is NOT a prodrug, with free drug circulating. This explains the transfusion difference.

BOARD TRAP

Ticagrelor is reversible and twice-daily; it also causes dyspnea and ventricular pauses, and its benefit is blunted by aspirin doses >100 mg. Pairing ticagrelor with high-dose aspirin is a trap.

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13. TTP — rare clopidogrel side effect

WHAT YOU SEE

Side effects strip listing bleeding, low cell counts, and TTP (thrombotic thrombocytopenic purpura), 'rare, life-threatening'.

WHAT IT ENCODES

Clopidogrel (and ticlopidine) can rarely cause thrombotic microangiopathy/TTP and neutropenia — ticlopidine being worst, which is why it was largely abandoned for clopidogrel.

BOARD TRAP

Neutropenia/agranulocytosis and TTP are the ticlopidine/clopidogrel class hazard. The board point: ticlopidine was replaced by clopidogrel because of higher hematologic toxicity.
